

Dakyung Kang

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EDUCATION

University of Utah

M.S. in Electrical and Computer Engineering

Salt Lake City, UT

Expected Aug 2028

University of Utah

B.S. in Electrical and Computer Engineering

Salt Lake City, UT

Dec 2025

PUBLICATION

H. Shin, **D. Kang**, J. Kim, and N. Kim, "CMOS Impedance Sensor with 128×128 Microelectrode Arrays for Measuring Surface Coverage of Microparticles," *2025 IEEE SENSORS Conference*, Vancouver, Canada, accepted for publication, Oct. 2025.

SKILLS

Programming Languages: C/C++, Python.

Hardware Description: Verilog, SystemVerilog, RTL Design.

EDA & Tools: Cadence (Innovus, Genus, Virtuoso), Calibre, MATLAB.

EXPERIENCE

Patent Engineer

Honesty & JR Partners

Seoul, Republic of Korea

Feb 2026 – Jun 2026

- Analyzed circuit operating principles of power semiconductor and display circuits at the transistor level based on inventor disclosure forms.
- Conducted prior art searches for 5 patent proposals across EV inverter circuits, MOSFET fabrication, identifying 3 reference patents per case and drafted patent specification and claims for display fabrication and display share/private mode circuits.

Lab Assistant

Biomedical Micro-Nano Systems Lab

Salt Lake City, UT

Aug 2024 – Dec 2025

- Validated sensing performance of **CMOS electrode array** by correlating impedance with optical microscopy across five microparticle concentrations.
- Measured impedance across electrode spacings ($3\mu\text{m}$ vs $4\mu\text{m}$, 20–100kHz), achieving 1.63x higher SNR with a low spacing gap and adopting it for subsequent immunoassay measurements.
- Analyzed $\Delta|Z|$ via heatmap visualization in **MATLAB** by mapping results to surface coverage in 0.25% to 98.65% and confirmed spatial correlation with microparticle distribution.

TECHNICAL PROJECTS

High Speed Data Acquisition Controller

Sep 2025 – Jun 2026

- Designed an 8-channel biosignal acquisition controller (ECG/EEG/EMG) in **SystemVerilog** for a bedside patient monitor, featuring Round-Robin, Priority, Weighted, and Dynamic arbitration to prioritize ECG sampling and a 16-entry FIFO with 2-level watermark status to decouple ADC sampling from SPI output.
- Built testbenches with 17 SystemVerilog Assertions and constrained-random stimulus covering FIFO overflow/underflow, simultaneous read/write, all 16 arbitration mode transitions, reset-during-operation, and all-8-channel contention, achieving 100% functional coverage via **Questa covergroups**.
- Integrated **8x4 Mac array** for real-time biosignal detection and tested against a **Python golden model** over 1,000 random inputs with 0 mismatches.

DES Processor – VLSI

Sep 2025 – Dec 2025

- Executed a full RTL-to-GDSII implementation flow for a 16-round Feistel DES engine using **Verilog** in a **TSMC 180nm** node.
- Resolved critical synthesis errors in **Genus** by redesigning tristate logic into synchronous register-based outputs and passed standard cell library compatibility.
- Achieved timing closure in **Innovus** by improving WNS from -12.742ns with 803 timing violations to +4.315ns through H-tree topology Clock Tree Synthesis.
- Ensured manufacturability with 0 DRC/LVS violations using **Calibre**, including automatic insertion of 17 antenna protection diodes via **Innovus** to resolve antenna effects.